
Variational Inference for Uncertain Optimal Transport via Sinkhorn Parametrization

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Abstract

Optimal Transport (OT) traditionally relies on a fixed ground cost to produce a single deterministic transport plan, a practice that overlooks the inherent variability and noise in real-world data. While recent sampling-based approaches to OT offer a principled way to quantify this uncertainty, they are computationally prohibitive and struggle to scale. In this paper, we introduce Sinkhorn-parameterized Variational Inference (SPVI), the first scalable variational framework for performing posterior inference over transport plans. Our key insight is that the Sinkhorn map can be treated as a differentiable reparameterization of the set of entropic plans. This enables the use of flexible generative models like normalizing flows to approximate distributions over transport plans while enforcing marginal constraints. We experimentally demonstrate that our method matches the quality of intensive sampling techniques at a fraction of the computational cost, scaling effectively to large-scale problems.

1. Introduction

Optimal transport (OT) is a powerful framework for comparing and transforming probability distributions, and has gained significant traction in machine learning for defining distributional distances and generative modeling metrics (Cuturi, 2013; Peyré & Cuturi, 2019). Classical OT assumes a fixed, known cost function (e.g. a ground metric between samples) and yields a single deterministic matching between points, or *transport plan*, that minimizes the total transportation cost. In many real-world scenarios, however, the cost

is not exact or deterministic and may be uncertain or randomly fluctuating. This is, for instance, the case in logistics (Agueh & Carlier, 2011), where transport costs might be affected by random traffic conditions; in single-cell biology (Schiebinger et al., 2019), where batch effects can perturb the computed costs; or when data are missing in tabular datasets (Bleistein et al., 2025).

When multiple realizations of the random matching process are available, taking advantage of this observed variability is crucial for downstream applications such as predictions made from the transport plan or assessing possible uncertainty about future matchings. Concretely, given novel samples to match, one might want to infer a *distribution* over possible transport plans rather than a single fixed plan. We therefore ask the following question:

How can we take advantage of multiple observed noisy transport plans to learn a distribution over future matchings?

As illustrated in Figure 1, traditional sampling methods can capture these distributions, but they are often computationally prohibitive for real-time applications. We tackle this problem by building upon advances in variational inference and deep generative modeling. Variational Inference (VI) (Hoffman et al., 2013; Ranganath et al., 2014) is a widely used approach to approximating complex posterior distributions by optimizing over a family of tractable distributions. A cornerstone of modern VI with continuous latent-variable models is the *reparameterization trick* (Kingma & Welling, 2022; Rezende et al., 2014), which enables low-variance gradient estimates by expressing random samples as deterministic functions of parameters and noise (as done in Variational Auto Encoders (VAE) (Kingma & Welling, 2019) and many Bayesian deep learning methods (Blundell et al., 2015; Gal & Ghahramani, 2016)). We bring this idea to OT by deriving an efficient reparameterization trick for the optimal transport problem, which in turn allows us to sample from a parametrized distribution over random plans.

In order to learn a flexible distribution over transport plans, we use normalizing flows. Normalizing flows (Rezende & Mohamed, 2015; Dinh et al., 2017) construct complex distributions by composing invertible transformations of a base

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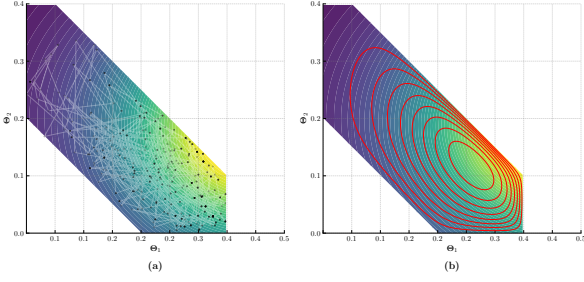


Figure 1. (a) 500 posterior samples on a 2×3 transport polytope generated via Hamiltonian Monte Carlo (HMC) with BayesOT. HMC suffers from high latency and poor mixing due to rigid marginal constraints. (b) SPVI (red contours) is fast and approximates the posterior with high fidelity, enabling efficient uncertainty quantification for optimal transport.

noise distribution. In principle, one could directly attempt to design a normalizing flow on the manifold of transport plans using a parameterization that guarantees the Jacobian preserves the polytope constraints. However, defining flows that exactly respect marginal constraints is highly non-trivial. Instead, our approach uses the Sinkhorn mapping (Knight, 2007), a non-invertible but structured iterative scaling procedure that transforms a strictly positive matrix into a coupling with specified marginals by alternating between row and column normalization until convergence. The unconstrained Sinkhorn map is non-injective because diagonal rescalings of the input kernel leave the resulting coupling unchanged. We remove this by working in a gauge-fixed log-kernel subspace, where the Sinkhorn map becomes a smooth global chart of the relative interior of the transport polytope, allowing us to combine flexible normalizing flows with exact marginal constraint satisfaction. The resulting variational family has a tractable density and can therefore be optimized using the ELBO (Huszár, 2017; Uppal et al., 2023).

Contributions. Our contributions are as follows.

1. We introduce **Sinkhorn-parameterized Variational Inference (SPVI)**, a variational framework for posterior inference over the transport polytope.
2. We prove that the truncated Sinkhorn operator and its associated Jacobian converge geometrically to their exact counterparts, justifying the use of finite-step iterations in the ELBO.
3. We empirically show the scalability of SPVI compared to sampling-based approaches.
4. We demonstrate the utility of our approach in tasks such as domain adaptation, joint representation and transport learning, and single-cell data analysis.

2. Related Work

Optimal Transport and Entropic Regularization. OT has become a cornerstone of modern machine learning and has been embraced as a means to compare distributions and define generative model objectives (Cuturi, 2013; Arjovsky et al., 2017; Peyré & Cuturi, 2019). Classical OT seeks a coupling Γ that minimizes the expected cost $\langle C, \Gamma \rangle$ on the transport polytope $\Pi(\mu, \nu)$ (defined in Sec. 3). Without additional regularization, the solution to the OT problem is often sparse and hard to compute at scale. Variants of *regularized* OT have since been proposed to alleviate this computational burden and enforce desirable structural properties, including quadratic, group-sparse, and Tsallis regularizations. We focus in particular on *entropy-regularized* OT, which adds a convex entropy penalty scaled by a regularization parameter $\varepsilon > 0$. This relaxation yields a smoother, unique optimal plan that can be solved efficiently via the Sinkhorn–Knopp matrix scaling algorithm. Entropic OT not only accelerates OT computations by orders of magnitude but also admits a probabilistic interpretation: it is equivalent to an entropy-minimization (Schrödinger bridge) problem (Léonard, 2014; Bortoli et al., 2021; Bunne et al., 2023a). Specifically, the optimal coupling is the KL projection of a Gibbs kernel $K_\varepsilon = \exp(-C/\varepsilon)$ onto $\Pi(\mu, \nu)$, representing the most likely coupling subject to marginal constraints (Cuturi, 2013; Léonard, 2014; Stromme, 2023).

While regularization provides a smoothed objective, it does *not* provide a full Bayesian treatment of uncertainty and the resulting solution is a point estimate that fails to quantify sensitivity to varying costs. Our work interprets the regularizer as a negative log-prior, transforming the transport problem from a single optimization task into a posterior inference task over the entire polytope.

Bayesian Optimal Transport and Uncertain Costs. Uncertain OT has been addressed by a few recent works. The BayesOT framework of Mallasto et al. (2021) is the first to explicitly infer a posterior over transport plans given a stochastic cost function. They define a prior distribution on the space of couplings and use a likelihood based on the cost observations, resulting in a posterior that can be sampled via Markov chain Monte Carlo. Their work demonstrated that one can obtain meaningful uncertainty quantification for OT, for example by producing credible intervals for entries of the transport matrix in a demographic matching problem, and that the maximum a posteriori (MAP) coincides with entropy-regularized solutions under appropriate priors. Nevertheless, a limitation of BayesOT is the computational overhead and the long mixing times of MCMC. In fact, earlier attempts at Bayesian inference on contingency tables (joint distributions) with fixed marginals, such as the work of Rosen et al. (2001), encounter convergence issues when using naive Gibbs samplers, precisely because of the rigid

marginal constraints. Frogner & Poggio (2019) addresses a related problem by proposing a family of models with a prior over joint distributions and deriving an *exact* MAP solution via Dijkstra’s algorithm. Their approach yields a point estimate of the coupling that maximizes the posterior, thus generalizing OT to incorporate prior knowledge without sacrificing the marginal constraints.

In essence, we blend the efficiency of the Sinkhorn updates and projections (as used by Frogner & Poggio for MAP inference) with a variational approximation that captures the posterior uncertainty as BayesOT intended. Recent studies in distributionally robust optimization (De et al., 2024; Blanchet et al., 2021; Mohajerin Esfahani & Kuhn, 2018; Mukherjee et al., 2021) and robust OT (Balaji et al., 2020) have also considered worst-case or adversarial cost perturbations (Sinha et al., 2018), but these typically yield a single robust coupling rather than a distribution of couplings.

Differentiable Optimal Transport. Our work crucially takes advantage of the differentiability of the Sinkhorn algorithm. This was also exploited by Gumbel–Sinkhorn networks (Mena et al., 2018), which approximate permutation matrices in a differentiable manner by adding Gumbel noise to a cost matrix and applying Sinkhorn normalization. In our case, we do not restrict to permutations, as our aim is to sample general couplings, but we similarly rely on the Sinkhorn algorithm’s ability to produce doubly stochastic matrices as outputs of a neural network layer and on its differentiability. Recent research has further incorporated Sinkhorn layers in neural architectures, for example as differentiable relaxations for ranking, sorting, and related permutation-based operators (Adams & Zemel, 2011; Cuturi et al., 2019; Tay et al., 2020), to learn flexible weight-sharing patterns via learnable doubly-stochastic matrices (van der Linden et al., 2024), or to enforce structural constraints by constraining learned mixing matrices to lie in the Birkhoff polytope via Sinkhorn–Knopp normalization (Yang & Gao, 2026).

3. Background on Optimal Transport

We recall the basic notions of OT and its regularized variants. We work throughout in the finite, discrete setting.

Discrete optimal transport. Let $\mu \in \Delta_{++}^{n-1}$ and $\nu \in \Delta_{++}^{m-1}$ be probability vectors on finite sets $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_m\}$, so that $\mu_i > 0$, $\sum_{i=1}^n \mu_i = 1$ and similarly for ν . We consider *cost functions*

$$c : \{x_1, \dots, x_n\} \times \{y_1, \dots, y_m\} \rightarrow \mathbb{R}_+$$

which give rise to nonnegative cost matrices $C \in \mathbb{R}_+^{n \times m}$ with entries $C_{ij} = c(x_i, y_j)$. A *transport plan* (or *coupling*) between μ and ν is a nonnegative matrix $\Gamma \in \mathbb{R}_+^{n \times m}$ whose row and column sums coincide with μ and ν :

$$\Pi(\mu, \nu) := \{\Gamma \in \mathbb{R}_+^{n \times m} : \Gamma \mathbf{1}_m = \mu, \Gamma^\top \mathbf{1}_n = \nu\}, \quad (1)$$

where $\mathbf{1}_k$ denotes the k -dimensional vector of ones. The quantity Γ_{ij} represents the amount of mass moved from x_i to y_j . The set $\Pi(\mu, \nu)$ is a convex polytope in $\mathbb{R}^{n \times m}$, usually referred to as the *transport polytope*.

Given a cost matrix C , the *Kantorovich optimal transport problem* seeks a coupling of minimal total cost:

$$\text{OT}(\mu, \nu; C) := \min_{\Gamma \in \Pi(\mu, \nu)} \langle C, \Gamma \rangle \quad (2)$$

where $\langle A, B \rangle = \sum_{ij} A_{ij} B_{ij}$ is the Frobenius inner product. Any minimizer $\Gamma^* \in \Pi(\mu, \nu)$ is called an *optimal transport plan*. When c arises from a metric $C_{ij} = d(x_i, y_j)^p$, the corresponding continuum analogue satisfies $\text{OT}(\mu, \nu; d^p) = \mathcal{W}_p(\mu, \nu)^p$, where $\mathcal{W}_p$ is the p -Wasserstein distance. In this paper we remain in the discrete setting and focus on the structure of $\Pi(\mu, \nu)$ and on the effect of cost uncertainty.

Geometry of the transport polytope. The transport polytope $\Pi(\mu, \nu)$ is the intersection of the non-negative orthant $\mathbb{R}_+^{n \times m}$ with an affine subspace determined by the marginal constraints. Let $\mathcal{A}$ denote this affine subspace of matrices with fixed row and column sums:

$$\mathcal{A} := \{\Gamma \in \mathbb{R}^{n \times m} : \Gamma \mathbf{1}_m = \mu, \Gamma^\top \mathbf{1}_n = \nu\}.$$

Thus, we can compactly write $\Pi(\mu, \nu) = \mathcal{A} \cap \mathbb{R}_+^{n \times m}$.

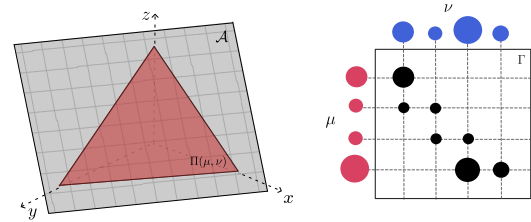


Figure 2. Geometric and matrix representations of the transport polytope. (Left) The transport polytope $\Pi(\mu, \nu)$ (red) is the intersection of the non-negative orthant with the affine subspace $\mathcal{A}$ defined by the marginal constraints. (Right) A discrete transport plan Γ visualized as a coupling matrix; the sizes of the black circles represent the mass values Γ_{ij} , while the marginals μ (red) and ν (blue) are obtained by summing the rows and columns, respectively.

While $\mathcal{A}$ is defined by $n + m$ linear equalities, the constraints are not independent due to the mass conservation $\sum_i \mu_i = \sum_j \nu_j = 1$. One constraint is redundant, making the codimension of $\mathcal{A}$ equal to $n + m - 1$. Consequently, $\dim(\mathcal{A}) := r = (n - 1)(m - 1)$. Since we assume strictly positive marginals, $\Pi(\mu, \nu)$ possesses a relative interior

$$\mathcal{M} := \{\Gamma \in \Pi(\mu, \nu) : \Gamma_{ij} > 0 \ \forall i, j\} \quad (3)$$

which is an open subset of $\mathcal{A}$ with $\dim(\mathcal{M}) = r$. As the Kantorovich problem (2) is a linear program, its solutions are guaranteed to exist and typically lie on this boundary,

yielding plans with at most $n + m - 1$ non-zero entries. Crucially, the vertices and facets of the boundary $\partial\Pi(\mu, \nu)$ represent sparse transport plans.

Regularized optimal transport. While mathematically elegant, this sparsity makes the solution highly sensitive to perturbations in the cost matrix C , typically non-differentiable with respect to parametrizations of the cost matrices, and computationally expensive to obtain for high-dimensional distributions. This motivates *regularized* OT problems of the form

$$\text{OT}_R(\mu, \nu; C) := \min_{\Gamma \in \Pi(\mu, \nu)} \{ \langle C, \Gamma \rangle + R(\Gamma) \}, \quad (4)$$

where $R : \mathbb{R}^{n \times m} \rightarrow \mathbb{R} \cup \{+\infty\}$ is a strictly convex regularizer. Under strict convexity of R on $\Pi(\mu, \nu)$, problem (4) admits a unique minimizer (Peyré & Cuturi, 2019), which we denote by

$$\Gamma_R^*(\mu, \nu; C) \in \Pi(\mu, \nu).$$

A particularly important choice is the (negative) entropy regularizer

$$H(\Gamma) := - \sum_{i,j} \Gamma_{ij} \log \Gamma_{ij}, \quad R(\Gamma) = -\varepsilon H(\Gamma), \quad (5)$$

with $\varepsilon > 0$. The corresponding problem

$$\text{OT}_\varepsilon(\mu, \nu; C) := \min_{\Gamma \in \Pi(\mu, \nu)} \{ \langle C, \Gamma \rangle - \varepsilon H(\Gamma) \} \quad (6)$$

is the now standard *entropic* optimal transport problem. Since the gradient of $H(\Gamma)$ diverges as any $\Gamma_{ij} \rightarrow 0$, the entropy pulls the solution away from the sparse boundary $\partial\Pi(\mu, \nu)$ and into the relative interior $\mathcal{M}$. Its unique solution Γ_ε^* can be computed using the classical Sinkhorn–Knopp algorithm (see Section 4) by iterative row/column scaling.

Gibbs posteriors over couplings. Suppose we observe a collection of cost matrices $\mathcal{C} = \{C^{(1)}, \dots, C^{(N)}\} \subset \mathbb{R}_+^{n \times m}$ and wish to quantify the uncertainty this induces in the transport plan. A natural object of interest is then a *distribution* over couplings $\Gamma \in \Pi(\mu, \nu)$ that combines prior structural preferences with information from the observed costs. A convenient way to encode such a distribution is via a *Gibbs measure* on $\Pi(\mu, \nu)$ with an energy functional $E(\Gamma)$. All densities are defined with respect to the r -dimensional Hausdorff measure $\mathcal{H}^r$ on the affine hull of the polytope. When $R(\Gamma) = -\log p(\Gamma \mid \mu, \nu)$, these measures correspond to Bayesian posteriors. This leads to two natural energy functionals corresponding to different posterior behaviors, as employed by Mallasto et al. (2021). The L_1 optimality notion uses the energy functional

$$E(\Gamma) := R(\Gamma) + \beta \sum_{k=1}^N \langle C^{(k)}, \Gamma \rangle,$$

where $\beta > 0$ is an inverse temperature parameter. The resulting density is

$$p(\Gamma \mid C^{(1:N)}, \mu, \nu) \propto \exp(-E(\Gamma)) \mathbf{1}_{\Pi(\mu, \nu)}$$

The MAP estimator for this posterior corresponds to the standard regularized OT solution:

$$\Gamma_{\text{MAP}} = \arg \min_{\Gamma \in \Pi(\mu, \nu)} \{ R(\Gamma) + \beta \langle \bar{C}, \Gamma \rangle \}$$

Because this depends only on the aggregate cost $\bar{C}$, it cannot capture multimodality arising from conflicting cost realizations. To capture regime-dependent structures, the L_2 construction uses a log-sum-exp aggregation:

$$E(\Gamma) := R(\Gamma) - \log \left(\sum_{k=1}^N \exp(-\beta \langle C^{(k)}, \Gamma \rangle) \right)$$

The associated density simplifies to a mixture-like form:

$$p(\Gamma \mid C^{(1:N)}) \propto \exp(-R(\Gamma)) \sum_{k=1}^N \exp(-\beta \langle C^{(k)}, \Gamma \rangle) \mathbf{1}_{\Pi(\mu, \nu)} \quad (7)$$

This allows the posterior to capture multimodality when different cost matrices favor distinct transport plans. BayesOT defines a likelihood via auxiliary binary optimality variables and performs posterior inference by HMC/NUTS sampling on a constrained chart (Brubaker et al., 2012) of the transport polytope to respect the marginal constraints, which becomes costly at scale (Beskos et al., 2011; Brooks et al., 2011), with multimodality further increasing autocorrelation (Woodard et al., 2009). Variational inference replaces this repeated constrained sampling with differentiable optimization of a reparameterized family.

4. Sinkhorn–parameterized variational inference

We approximate the Gibbs posterior (7) over couplings $\Gamma \in \Pi(\mu, \nu)$ by a reparameterized variational family supported on $\mathcal{M}$.

Definition 4.1. (Sinkhorn, 1964) *The Sinkhorn operator $\mathcal{S} : \mathbb{R}_+^{n \times m} \times \Delta_{++}^{n-1} \times \Delta_{++}^{m-1} \rightarrow \mathcal{M}$ maps a strictly positive kernel K to the unique matrix $\Gamma = \text{diag}(u)K\text{diag}(v)$ in the transport polytope $\Pi(\mu, \nu)$. The scaling vectors $u \in \mathbb{R}_{++}^n$ and $v \in \mathbb{R}_{++}^m$ are defined as the unique (up to a scalar factor) fixed points of the alternating scaling steps:*

$$u \leftarrow \frac{\mu}{Kv} \quad \text{and} \quad v \leftarrow \frac{\nu}{K^\top u}, \quad (8)$$

where the division is performed element-wise.

To construct a valid density on $\mathcal{M}$, we must define a map that naturally enforces the marginal constraints while handling the over-parameterization of the logit space.

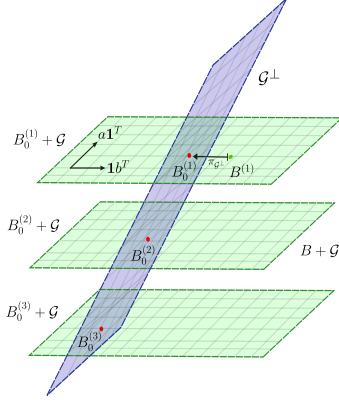


Figure 3. The Sinkhorn map is invariant to additive row/column offsets in log space: $B \mapsto B + a\mathbf{1}_m^\top + \mathbf{1}_n b^\top$, which generates a gauge orbit $B + \mathcal{G}$ shown as a family of parallel green planes. The blue plane denotes the gauge slice $\mathcal{G}^\perp$. Each orbit intersects $\mathcal{G}^\perp$ in exactly one point $B_0^{(k)}$, providing a unique gauge-fixed representative. The arrow is the projection $B^{(1)} \xrightarrow{\pi_{\mathcal{G}^\perp}} B_0^{(1)}$.

Gauge-fixed logit embedding. The Sinkhorn operator $\mathcal{S}(K; \mu, \nu)$ is invariant to diagonal rescaling of the kernel, meaning $\mathcal{S}(K) = \mathcal{S}(D_1 K D_2)$ for any diagonal matrices D_1, D_2 with positive entries. In the log-domain, where $B = \log K$, this invariance takes the form of additive gauge freedom: adding a row-wise or column-wise constant to B does not change the resulting coupling. We define the subspace of these redundant additive perturbations as

$$\mathcal{G} := \{a\mathbf{1}_m^\top + \mathbf{1}_n b^\top : a \in \mathbb{R}^n, b \in \mathbb{R}^m\}.$$

The pre-image of any single coupling $\Gamma \in \mathcal{M}$ under the log-Sinkhorn map is an affine subspace of the form $B + \mathcal{G}$. As illustrated in Figure 3, to ensure bijectivity, we restrict our logits to the orthogonal complement $\mathcal{G}^\perp = \{B \in \mathbb{R}^{n \times m} : B\mathbf{1}_m = 0, B^\top \mathbf{1}_n = 0\}$. This choice is without loss of generality, as every equivalence class in the logit space modulo $\mathcal{G}$ has exactly one representative in $\mathcal{G}^\perp$.

Proposition 4.2. Let $\mathcal{G} = \{a\mathbf{1}_m^\top + \mathbf{1}_n b^\top : a \in \mathbb{R}^n, b \in \mathbb{R}^m\}$ be the subspace of additive gauge freedoms, and let $\mathcal{G}^\perp$ be its orthogonal complement in $\mathbb{R}^{n \times m}$. Let $Q \in \mathbb{R}^{nm \times r}$ be a matrix with orthonormal columns spanning $\text{vec}(\mathcal{G}^\perp)$. Define the map $\Psi : \mathbb{R}^r \rightarrow \mathcal{M}$ by

$$\Psi(z) := \mathcal{S}(\exp(\text{mat}(Qz)); \mu, \nu) \quad (9)$$

where $\text{mat}(\cdot)$ is the matricization operator that reshapes a vector in $\mathbb{R}^{nm}$ into a matrix $\mathbb{R}^{n \times m}$. Then Ψ is a smooth global diffeomorphism from $\mathbb{R}^r$ onto $\mathcal{M}$. In particular, it is globally injective and its Jacobian $J_\Psi(z)$ has full column rank r everywhere.

Proof. See Appendix D. $\square$

Sampler. We define the reparameterized sampler Γ via the following sequence of transformations. We first sample $\xi \sim p(\xi) = \mathcal{N}(0, I_r)$, and apply an invertible normalizing flow $f_\phi : \mathbb{R}^r \rightarrow \mathbb{R}^r$, conditioned on $\mathcal{C}$ with tractable Jacobian determinant. To ensure numerical stability, we apply a bounded squashing map

$$s(x) = c \tanh(x/c)$$

to obtain a latent code $z = h_\phi(\xi) = s(f_\phi(\xi)) \in (-c, c)^r$. We then map z to the gauge-fixed logit space via Q as

$$B(z) := \text{mat}(Qz) \in \mathcal{G}^\perp.$$

Finally, we exponentiate to obtain the kernel $K = \exp(B(z)) \in \mathbb{R}_{++}^{n \times m}$ and apply the Sinkhorn scaling operator $\mathcal{S}(\cdot; \mu, \nu)$ to enforce marginals:

$$\Gamma(\xi) := \mathcal{S}(\exp(B(h_\phi(\xi))); \mu, \nu) \in \mathcal{M}. \quad (10)$$

In practice, $\mathcal{S}$ is implemented via a fixed number of iterations T (see Appendix C) chosen so that the marginal violation is below a prescribed tolerance. We give a description of our sampling procedure in Algorithm 1.

4.1. Variational objective and density on $\mathcal{M}$

We optimize the variational parameters ϕ by maximizing the ELBO. Let $\tilde{F}(z) := \mathcal{S}(\exp(\text{mat}(Qz)); \mu, \nu)$ denote the map from the squashed latent code to the coupling. The density of the pushforward distribution q_ϕ on the manifold $\mathcal{M}$ is obtained via the area formula.

Because $\tilde{F}$ is a globally injective immersion, the area formula applies without summation over preimages. The pushforward density q_ϕ with respect to $\mathcal{H}^r$ on $\mathcal{M}$ is:

$$\log q_\phi(\Gamma | \mathcal{C}) = \log p(\xi) - \log |\det J_{h_\phi}(\xi)| - \frac{1}{2} \log \det(G(z))$$

The term $G(z) = J_{\tilde{F}}(z)^\top J_{\tilde{F}}(z)$ is the Gram matrix and the factor $\sqrt{\det G(z)}$ measures the local volume expansion induced by the embedding. Since $\tilde{F}$ is an immersion, $J_{\tilde{F}}$ has full rank r , ensuring the Gram matrix is positive definite.

Substituting this density into the standard ELBO definition, we obtain the tractable objective (up to an additive constant independent of ϕ):

$$\mathcal{L}(\phi) \propto \mathbb{E}_{\xi \sim \mathcal{N}(0, I_r)} \left[-E(\Gamma(\xi)) + \log |\det J_{h_\phi}(\xi)| + \frac{1}{2} \log \det G(z) \right]. \quad (11)$$

We use S Monte Carlo samples of ξ to approximate the expectation cheaply. Since $\log \det G(z) = \text{Tr}(\log G(z))$, we can use the Hutchinson trace estimator (Hutchinson,

Algorithm 1 Sinkhorn-parameterized Variational Inference

Input: marginals μ, ν ; costs C ; energy functional $E(\cdot)$; flow f_ϕ ; gauge basis Q ; Sinkhorn iterations T ; MC batch size S ; Hutchinson probes M ; Lanczos steps L ; step η

Output: Variational parameters ϕ

Initialize ϕ

repeat

 Initialize $\hat{\mathcal{L}} \leftarrow 0$

for $s = 1$ **to** S **do**

 Sample $\xi^{(s)} \sim \mathcal{N}(0, I_r)$

$z^{(s)} \leftarrow s(f_\phi(\xi^{(s)}; C))$

$\Gamma^{(s)} \leftarrow \mathcal{S}(\exp(\text{mat}(Qz^{(s)})); \mu, \nu, T)$

$E^{(s)} \leftarrow E(\Gamma^{(s)})$

$\Delta_{ld}^{(s)} \leftarrow 0$

for $m = 1$ **to** M **do**

 Sample probe $\omega^{(m)} \in \{-1, 1\}^r$

$v^{(0)} \leftarrow \omega^{(m)} / \|\omega^{(m)}\|$

$T_L^{(s,m)} \leftarrow L$ step Lanczos on $G(z^{(s)})$ with $v^{(0)}$

$\Delta_{ld}^{(s)} \leftarrow \Delta_{ld}^{(s)} + \|\omega^{(m)}\|^2 (e_1^\top \log(T_L^{(s,m)}) e_1)$

end for

$\log \det \leftarrow \Delta_{ld}^{(s)} / M$

$\ell^{(s)} \leftarrow -E^{(s)} + \log |\det J_{h_\phi}(\xi^{(s)})| + \frac{1}{2} \widehat{\log \det}$

$\hat{\mathcal{L}} \leftarrow \hat{\mathcal{L}} + \ell^{(s)}$

end for

$\phi \leftarrow \phi + \eta \nabla_\phi(\hat{\mathcal{L}}/S)$

until convergence

1989). We draw M probe vectors $(\omega^{(1)}, \dots, \omega^{(M)} \in \mathbb{R}^r)$ with $\mathbb{E}[\omega\omega^\top] = I$ (Rademacher (± 1) or standard Gaussian), and use the approximation

$$\text{Tr}(\log G(z)) \approx \frac{1}{M} \sum_{i=1}^M (\omega^{(i)})^\top \log(G(z)) \omega^{(i)}.$$

Finally, for each quadratic form, we use Stochastic Lanczos Quadrature (SLQ) (Ubaru et al., 2017; Chen et al., 2021). For each probe $\omega^{(i)}$, we run L steps of Lanczos on $G(z)$ with starting vector

$$q_1^{(i)} = \frac{\omega^{(i)}}{\|\omega^{(i)}\|}.$$

This produces an $L \times L$ symmetric tridiagonal matrix $T_L^{(i)}(z)$. SLQ approximates

$$(\omega^{(i)})^\top \log(G(z)) \omega^{(i)} \approx \|\omega^{(i)}\|^2 e_1^\top \log(T_L^{(i)}(z)) e_1,$$

where $(e_1 = (1, 0, \dots, 0)^\top \in \mathbb{R}^L)$. This requires only matrix-vector products with the Jacobian $J_{\hat{F}}$, which are efficiently computed via automatic differentiation. We prove in Appendix A that the coupling map and its associated Jacobian converge geometrically to their exact counterparts as $T \rightarrow \infty$. The time complexity scales with the size $n \times m$ as $\mathcal{O}(SnmTML)$.

5. Experiments

To ensure reproducibility, all architectural details, hyperparameters, and optimization settings are given in Appendix B.

5.1. Domain Adaptation on Toy Datasets

We evaluate SPVI on a cross-domain transport task between MNIST (source) and USPS (target), a benchmark setting in domain adaptation (DA) where OT is frequently employed to align distributions across varying acquisition domains (Courtney et al., 2015; Redko et al., 2019).

5.1.1. ALEATORIC UNCERTAINTY MODELING

To study uncertainty stemming from intrinsic morphological variation, we analyze four semantically ambiguous pairs: $\{4 \rightarrow 9\}$, $\{3 \rightarrow 8\}$, $\{3 \rightarrow 7\}$, and $\{5 \rightarrow 9\}$. For any coupling $\Gamma \in \Pi(\mu, \nu)$, we define the morphed image via the barycentric projection $\mathcal{T}(\Gamma) \in \mathbb{R}^d$:

$$\mathcal{T}(\Gamma) := \text{diag}(\mu)^{-1} \Gamma \mathbf{Y}$$

where $\mathbf{Y} \in \mathbb{R}^{m \times d}$ is the matrix of target images, allowing us to visualize transport uncertainty directly for a source i , as shown in Figure 4. If Γ maps source i to target j , the resulting morph is the target image y_j . If Γ splits the mass of i across several targets, the morph is their weighted average.

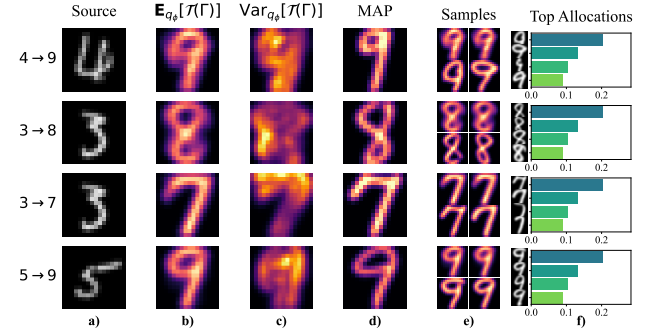


Figure 4. Each row shows transport from (a) a single MNIST source digit to USPS target digits. (b) Expected morph $\mathbb{E}_{q_\phi}[\mathcal{T}_i(\Gamma)]$, where $\mathcal{T}_i(\Gamma)$ is the i -th row of $\mathcal{T}(\Gamma)$; structural blur indicates regions of transport uncertainty. (c) Pixel-wise variance $\text{Var}_{q_\phi}[\mathcal{T}_i(\Gamma)]$, where brighter colors indicate higher transport ambiguity. (d) MAP on the mean cost (deterministic baseline). (e) Individual posterior samples. (f) Top four target indices by posterior mean mass.

5.1.2. MULTIMODALITY AND MIXED TARGETS

A key challenge in OT is when a single source mass can be plausibly mapped to distinct target clusters. We test our model’s ability to resolve such ambiguities using three mixed target sets: $\{1\} \rightarrow \{3, 6\}$, $\{0\} \rightarrow \{2, 5, 9\}$, and $\{3\} \rightarrow \{4, 8\}$. As shown in Figure 5, standard MAP on the

mean cost selects only the lowest-cost mode. In contrast, our VI approach identifies distinct posterior modes and the posterior samples reflect the true diversity of potential mappings. These results demonstrate that SPVI identifies not

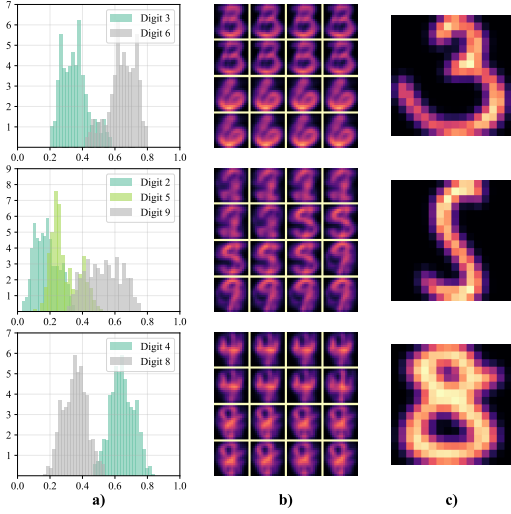


Figure 5. (a) Transport mass distribution across target digit classes shows multimodal posterior. (b) Posterior samples exhibit a diverse set of morphings favoring different digits. (c) Barycentric projection of the MAP on the mean cost (deterministic baseline).

just the most likely transport plan, but also the plausible alternatives. We now demonstrate the scalability of our framework in comparison to the sampling-based BayesOT.

5.2. High-Dimensional Scalability

We scale the problem from $m = n = 32$ to $n = 512$ using two Gaussian point clouds in $\mathbb{R}^2$ with squared Euclidean costs. We simulate uncertainty via $N = 20$ positive noisy realizations obtained from multiplicative log-normal perturbations of the base cost. We use a matrix-valued affine coupling flow, i.e. a RealNVP-style coupling architecture applied to the gauge-fixed logit matrix, using structured masks and convolutional conditioner networks (Dinh et al., 2015; Kingma & Dhariwal, 2018; Negri et al., 2023). As a gold-standard reference, we employ a HMC sampler (Malasto et al., 2021) using the No-U-Turn Sampler (NUTS) (Hoffman & Gelman, 2014), and also add a comparison of MAP on the mean cost. Table 1 reports wall-clock runtime across scales. Posterior distance and cost-variance diagnostics are reported in Appendix B.2.

5.3. Joint Representation and Transport Learning

In many practical applications, the cost matrix is computed from learned features from raw data, and different feature extractors provide divergent views of the data. Consider a domain adaptation setting on the Office-31 dataset (Saenko et al., 2010) with a labeled source $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^n$ and

Table 1. Runtime benchmark for Sinkhorn-VI against HMC across scales ($N = 20$ cost samples). MAP runtimes measure Sinkhorn on the mean cost with 100 iterations over 10 timing repeats. HMC runtimes for $n \leq 256$ use scale-dependent NUTS budgets (50 burn-in/100 samples at $n = 32$, 100/200 at $n = 64$, 200/500 at $n = 128$, and 500/1000 at $n = 256$) and are averaged over 10 runs; SPVI results are averaged over 5 runs. HMC chains are monitored via $\hat{R} < 1.1$ with a 10-hour wall clock and do not converge for $n > 256$. All experiments are run on CPU.

Size (n)	Dim (r)	Method	Runtime (s) ↓
$n = 32$	$r = 961$	MAP	0.005 ± 0.001
		HMC	61.6 ± 23.0
		SPVI	59 ± 8
$n = 64$	$r = 3,969$	MAP	0.025 ± 0.011
		HMC	329.0 ± 34.2
		SPVI	157 ± 32
$n = 128$	$r = 16,129$	MAP	0.032 ± 0.007
		HMC	$1,217.1 \pm 238.9$
		SPVI	580 ± 116
$n = 256$	$r = 65,025$	MAP	0.069 ± 0.007
		HMC	$21,533 \pm 3,200$
		SPVI	$2,144 \pm 429$
$n = 512$	$r = 261,121$	MAP	0.161 ± 0.019
		HMC	—
		SPVI	$7,933 \pm 1,587$

an unlabeled target $\mathcal{D}_t = \{x_j^t\}_{j=1}^m$, where y_i^s represents the class label among $K = 31$ possible classes. Define a classifier $g_\psi : \mathcal{X} \rightarrow \mathbb{R}^K$, and denote $g_{\psi,c}(x)$ as the logit for class c . We employ 4 frozen encoders: ResNet-50 (He et al., 2016), DeiT (Touvron et al., 2021), DINOv2 (Oquab et al., 2024), and OpenCLIP (Cherti et al., 2023) to construct cost matrices $\{C^{(k)}\}_{k=1}^4$ as

$$C_{ij}^{(k)}(\psi) = \alpha \tilde{D}_{ij}^{(k)} + \eta(t) L_{ij}(\psi)$$

with a tunable α . Here $\tilde{D}_{ij}^{(k)}$ represents standardized ℓ_2 -normalized feature distances scaled by the matrix median to ensure that no view dominates purely by scale. The second term $L_{ij}(\psi)$ enforces label consistency using a one-vs-all squared hinge loss $\ell_{\text{sh}}(y, s) = \max(0, 1 - ys)^2$:

$$L_{ij}(\psi) = \sum_{c=1}^K \ell_{\text{sh}}(Y_{ic}^s, g_{\psi,c}(x_j^t)),$$

where $Y^s \in \{+1, -1\}^{n \times K}$ are the signed source labels. We optimize via block coordinate updates $(\phi^{(t)}, \psi^{(t)})$ over T_{outer} iterations.

- Update Transport (ϕ): Given classifier $\psi^{(t)}$, update variational parameters ϕ to maximize $\mathcal{L}(\phi; \mathcal{C}(\psi^{(t)}))$.
- Update Classifier (ψ): Minimize the posterior-predictive risk. Let $\hat{P}(\Gamma) = \Gamma^\top P^s \in \mathbb{R}^{m \times K}$ be

the soft target labels induced by the coupling (where $P^s \in \{0, 1\}^{n \times K}$ are one-hot source labels). We minimize:

$$\mathcal{J}(\psi) = \mathbb{E}_{\Gamma \sim q_\phi} \left[\sum_{j=1}^m \sum_{c=1}^K \mathcal{L}_{\text{target}}(j, c, \psi, \Gamma) \right] + \lambda_s \mathcal{L}_{\text{source}}(\psi) + \lambda_\Omega \|\psi\|_2^2$$

where the target loss weighs positive and negative predictions by the transported mass:

$$\mathcal{L}_{\text{target}}(j, c, \psi, \Gamma) = \hat{P}_{jc}(\Gamma) \ell_{\text{sh}}(+1, g_{\psi,c}(x_j^t)) + (1 - \hat{P}_{jc}(\Gamma)) \ell_{\text{sh}}(-1, g_{\psi,c}(x_j^t)) \quad (12)$$

and the source loss

$$\mathcal{L}_{\text{source}}(\psi) = \sum_{i=1}^n \sum_{c=1}^K \ell_{\text{sh}}(Y_{ic}^s, g_{\psi,c}(x_i^s)). \quad (13)$$

To avoid early collapse, $\eta(t)$ linearly ramps from 0 to 1 over the outer iterations, gradually introducing the label constraint as the classifier improves. We compare our method against SurK (Zhang et al., 2013), a surrogate-kernel approach; OT-MM, which applies optimal transport with class-wise regularization optimized via a majorization-minimization algorithm (Chapel et al., 2021); and JDOT, which performs joint distribution optimal transport for domain adaptation (Courty et al., 2017). We also report a baseline (Base) obtained by training on the source domain and evaluating directly on the target domain. Since SPVI provides a distribution, we compute the MAP transport plan to compare the accuracy in Table 2. This demonstrates that,

Table 2. Classification accuracy (%) on Office-31 tasks.

Domains	Base	SurK	OT-MM	JDOT	SPVI
amazon $\rightarrow$ webcam	79.85	81.56	85.96	84.05	84.02
amazon $\rightarrow$ dslr	86.21	87.67	79.21	87.48	86.81
webcam $\rightarrow$ amazon	79.79	78.48	90.57	<u>90.21</u>	90.01
webcam $\rightarrow$ dslr	96.56	95.32	99.99	<u>98.55</u>	97.43
dslr $\rightarrow$ amazon	83.91	82.27	90.59	88.59	<u>89.47</u>
dslr $\rightarrow$ webcam	96.74	92.31	<u>96.56</u>	96.17	96.88
Mean	87.18	86.27	90.48	90.84	<u>90.77</u>

despite being designed to model a distribution over transport plans, SPVI achieves performance comparable to standard methods that estimate a single deterministic transport plan.

5.4. Single-Cell Fate Dynamics

Optimal transport has received considerable attention in recent years in single-cell biology, where it is used to map cell populations during differentiation measured by destructive experiments that prohibit one-to-one matching of cells (Schiebinger et al., 2019; Bunne et al., 2023b; 2024). We study fate uncertainty during neuronal differentiation in the

Dentate Gyrus dataset for mouse hippocampus neurogenesis (Hochgerner et al., 2018). We treat the differentiation process as a cross-population transport problem between an early (source) and a late (target) cell population. Let $X \in \mathbb{R}^{n \times 30}$ and $Y \in \mathbb{R}^{m \times 30}$ denote the PCA embeddings of source and target cells, with rows $x_i^\top$ and $y_j^\top$ and uniform marginals, where $n = m = 300$. Interpreting source and target populations as snapshots of a biological process at two latent times, the coupling Γ induces a probabilistic transition rule. We interpret the conditional distribution $\pi(j | i) := \Gamma_{ij} / \mu_i$ as a discrete approximation of the transition density $p(y | x_i)$ of a latent stochastic evolution operator (Zhang et al., 2025; Gu et al., 2025; Lavenant et al., 2024). We propagate uncertainty from the inferred developmental dynamics by constructing an ensemble of plausible transition cost matrices $\mathcal{C}$ by repeatedly reconstructing the cell-state geometry and the forward developmental transition structure under bootstrap perturbations of the single-cell data. Each cost matrix represents one plausible realization of the latent developmental dynamics between the source and target populations. See Appendix B.4 for details.

Our variational approximation $q_\phi(\Gamma | \mathcal{C})$ represents the uncertainty over this transition law. We capture this uncertainty through the posterior predictive variance, or *plasticity*, defined as

$$V_i := \sum_{\ell=1}^{30} \text{Var}_{q_\phi}([\mathcal{T}_i(\Gamma)]_\ell)$$

which serves as a proxy for branching uncertainty. As shown in Figure 6, multipotent progenitor cells exhibit higher Shannon entropy in their posterior mean fate distributions compared to committed lineages. Furthermore, plotting V_i against developmental maturity reveals a funneling effect: early cells have less stable predicted destinations across plausible transition dynamics, whereas mature cells are transported consistently to a narrower region of the target population.

6. Conclusion and Future work

We have introduced a theoretically grounded framework for variational inference over the transport polytope. This allows us to learn a posterior distribution over possible transport plans and to assess uncertainty in matching on new samples. SPVI shows promising results on a large variety of tasks, including uncertainty-aware inference of single-cell fate dynamics, domain adaptation, and joint representation and transport learning.

SPVI is limited to balanced entropic optimal transport; extensions to unbalanced optimal transport would be highly valuable. Obtaining theoretical results on the optimization process under simple parametrizations of the normalizing flows would also be interesting. Lastly, we believe that

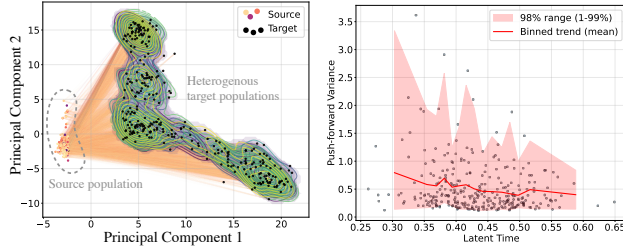


Figure 6. Left: Cells are mapped in PCA space. Source cells are colored by the Shannon entropy of the conditional fate distribution $\pi(j | i)$ derived from the posterior mean coupling. Transport lines are colored proportionally to their probability, connecting source cells to likely destinations. For the 5 most uncertain cells, density contours represent the variational posterior of barycentric fate mappings. Right: Relationship between V_i and developmental maturity (latent time). Variance is computed across 200 variational posterior samples. The red trend line indicates mean variance across 15 quantile bins, with shaded bands showing the 1st to 99th percentile range.

further developments of generative methods for optimal transport plans that can reflect fundamental randomness in matching mechanisms are of high interest.

Impact Statement

This paper introduces a scalable variational inference framework for Bayesian optimal transport that places expressive, reparameterized distributions directly on the interior of the transport polytope. By combining a gauge-fixed Sinkhorn parameterization with normalizing flows and a principled manifold density correction, our method replaces costly constrained MCMC with fast, differentiable optimization while retaining the ability to represent multimodal posteriors induced by uncertain or heterogeneous cost matrices. The approach enables uncertainty-aware transport plans at problem sizes where sampling-based BayesOT becomes impractical, and provides a general template for probabilistic inference over constrained couplings that can benefit downstream tasks such as robust matching, domain adaptation, and uncertainty-calibrated decision making.

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References

Adams, R. P. and Zemel, R. S. Ranking via sinkhorn propagation, 2011. URL <https://arxiv.org/abs/1106.1925>.

Agueh, M. and Carlier, G. Barycenters in the wasserstein space. *SIAM Journal on Mathematical Analysis*, 43(2): 904–924, 2011. doi: 10.1137/100805741. URL <https://doi.org/10.1137/100805741>.

Arjovsky, M., Chintala, S., and Bottou, L. Wasserstein generative adversarial networks. In Precup, D. and Teh, Y. W. (eds.), *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pp. 214–223. PMLR, 06–11 Aug 2017. URL <https://proceedings.mlr.press/v70/arjovsky17a.html>.

Balaji, Y., Chellappa, R., and Feizi, S. Robust optimal transport with applications in generative modeling and domain adaptation. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H. (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 12934–12944. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/9719a00ed0c5709d80dfef33795dcef3-Paper.pdf.

Beskos, A., Pinski, F., Sanz-Serna, J., and Stuart, A. Hybrid monte carlo on hilbert spaces. *Stochastic Processes and their Applications*, 121(10):2201–2230, 2011. ISSN 0304-4149. doi: <https://doi.org/10.1016/j.spa.2011.06.003>. URL <https://www.sciencedirect.com/science/article/pii/S0304414911001396>.

Blanchet, J., Murthy, K., and Zhang, F. Optimal transport based distributionally robust optimization: Structural properties and iterative schemes, 2021. URL <https://arxiv.org/abs/1810.02403>.

Bleistein, L., Bellet, A., and Josse, J. Optimal transport with heterogeneously missing data. *arXiv preprint arXiv:2505.17291*, 2025.

Blundell, C., Cornebise, J., Kavukcuoglu, K., and Wierstra, D. Weight uncertainty in neural network. In Bach, F. and Blei, D. (eds.), *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 1613–1622, Lille, France, 07–09 Jul 2015. PMLR. URL <https://proceedings.mlr.press/v37/blundell15.html>.

Bortoli, V. D., Thornton, J., Heng, J., and Doucet, A. Diffusion schrödinger bridge with applications to score-based generative modeling. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W. (eds.), *Advances in Neural Information Processing Systems*, 2021. URL <https://openreview.net/forum?id=9BnCwiXB0ty>.

- Brooks, S., Gelman, A., Jones, G., and Meng, X. *Handbook of Markov Chain Monte Carlo*. Chapman & Hall/CRC Handbooks of Modern Statistical Methods. CRC Press, 2011. ISBN 9781420079425. URL <https://books.google.de/books?id=qfRsAIKZ4rIC>.
- Brubaker, M., Salzmann, M., and Urtasun, R. A family of mcmc methods on implicitly defined manifolds. In Lawrence, N. D. and Girolami, M. (eds.), *Proceedings of the Fifteenth International Conference on Artificial Intelligence and Statistics*, volume 22 of *Proceedings of Machine Learning Research*, pp. 161–172, La Palma, Canary Islands, 21–23 Apr 2012. PMLR. URL <https://proceedings.mlr.press/v22/brubaker12.html>.
- Bunne, C., Hsieh, Y.-P., Cuturi, M., and Krause, A. The schrödinger bridge between gaussian measures has a closed form. In Ruiz, F., Dy, J., and van de Meent, J.-W. (eds.), *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 5802–5833. PMLR, 25–27 Apr 2023a. URL <https://proceedings.mlr.press/v206/bunne23a.html>.
- Bunne, C., Stark, S. G., Gut, G., Del Castillo, J. S., Levesque, M., Lehmann, K.-V., Pelkmans, L., Krause, A., and Rätsch, G. Learning single-cell perturbation responses using neural optimal transport. *Nature methods*, 20(11):1759–1768, 2023b.
- Bunne, C., Schiebinger, G., Krause, A., Regev, A., and Cuturi, M. Optimal transport for single-cell and spatial omics. *Nature Reviews Methods Primers*, 4(1):58, 2024.
- Chapel, L., Flamary, R., Wu, H., Févotte, C., and Gasso, G. Unbalanced optimal transport through non-negative penalized linear regression. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W. (eds.), *Advances in Neural Information Processing Systems*, volume 34, pp. 23270–23282. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/c3c617a9b80b3ae1ebd868b0017cc349-Paper.pdf.
- Chen, T., Trogon, T., and Ubaru, S. Analysis of stochastic lanczos quadrature for spectrum approximation. In Meila, M. and Zhang, T. (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 1728–1739. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/chen21s.html>.
- Cherti, M., Beaumont, R., Wightman, R., Wortsman, M., Ilharco, G., Gordon, C., Schuhmann, C., Schmidt, L., and Jitsev, J. Reproducible scaling laws for contrastive language-image learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2818–2829, 2023.
- Courty, N., Flamary, R., Tuia, D., and Rakotomamonjy, A. Optimal transport for domain adaptation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PP, 07 2015. doi: 10.1109/TPAMI.2016.2615921.
- Courty, N., Flamary, R., Habrard, A., and Rakotomamonjy, A. Joint distribution optimal transportation for domain adaptation. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/0070d23b06b1486a538c0eaa45dd167a-Paper.pdf.
- Cuturi, M. Sinkhorn distances: Lightspeed computation of optimal transport. In Burges, C., Bottou, L., Welling, M., Ghahramani, Z., and Weinberger, K. (eds.), *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc., 2013. URL https://proceedings.neurips.cc/paper_files/paper/2013/file/af21d0c97db2e27e13572cbf59eb343d-Paper.pdf.
- Cuturi, M., Teboul, O., and Vert, J.-P. Differentiable ranking and sorting using optimal transport. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/file/d8c24ca8f23c562a5600876ca2a550ce-Paper.pdf.
- De, A., Thielmann, A. F., and Säfken, B. NODE-GAMLSS: Interpretable uncertainty modelling via deep distributional regression. In *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, 2024. URL <https://openreview.net/forum?id=sWvTLlRkzy>.
- Dinh, L., Krueger, D., and Bengio, Y. Nice: Non-linear independent components estimation, 2015. URL <https://arxiv.org/abs/1410.8516>.
- Dinh, L., Sohl-Dickstein, J., and Bengio, S. Density estimation using real NVP. In *International Conference on Learning Representations*, 2017. URL <https://openreview.net/forum?id=HkpbN91x>.

- Franklin, J. and Lorenz, J. On the scaling of multidimensional matrices. *Linear Algebra and its Applications*, 114-115:717–735, 1989. ISSN 0024-3795. doi: [https://doi.org/10.1016/0024-3795\(89\)90490-4](https://doi.org/10.1016/0024-3795(89)90490-4). URL <https://www.sciencedirect.com/science/article/pii/0024379589904904>. Special Issue Dedicated to Alan J. Hoffman.
- Frogner, C. and Poggio, T. Fast and flexible inference of joint distributions from their marginals. In Chaudhuri, K. and Salakhutdinov, R. (eds.), *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 2002–2011. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/frogner19a.html>.
- Gal, Y. and Ghahramani, Z. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In Balcan, M. F. and Weinberger, K. Q. (eds.), *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pp. 1050–1059, New York, New York, USA, 20–22 Jun 2016. PMLR. URL <https://proceedings.mlr.press/v48/gal16.html>.
- Gu, A., Chien, E., and Greenewald, K. Partially observed trajectory inference using optimal transport and a dynamics prior. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=H8hO3T3DYe>.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. pp. 770–778, 06 2016. doi: 10.1109/CVPR.2016.90.
- Hochgerner, H., Zeisel, A., Lönnerberg, P., and Linnarsson, S. Conserved properties of dentate gyrus neurogenesis across postnatal development revealed by single-cell RNA sequencing. *Nature Neuroscience*, 21(2):290–299, 2018. doi: 10.1038/s41593-017-0056-2. URL <https://doi.org/10.1038/s41593-017-0056-2>.
- Hoffman, M. D. and Gelman, A. The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo. *Journal of Machine Learning Research*, 15(47):1593–1623, 2014. URL <http://jmlr.org/papers/v15/hoffman14a.html>.
- Hoffman, M. D., Blei, D. M., Wang, C., and Paisley, J. Stochastic variational inference. *Journal of Machine Learning Research*, 14(4):1303–1347, 2013. URL <http://jmlr.org/papers/v14/hoffman13a.html>.
- Huszár, F. Variational inference using implicit distributions, 2017. URL <https://arxiv.org/abs/1702.08235>.
- Hutchinson, M. A stochastic estimator of the trace of the influence matrix for laplacian smoothing splines. *Communication in Statistics- Simulation and Computation*, 18: 1059–1076, 01 1989. doi: 10.1080/03610919008812866.
- Idel, M. A review of matrix scaling and sinkhorn’s normal form for matrices and positive maps, 2016. URL <https://arxiv.org/abs/1609.06349>.
- Kingma, D. P. and Dhariwal, P. Glow: Generative flow with invertible 1x1 convolutions. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/file/d139db6a236200b21cc7f752979132d0-Paper.pdf.
- Kingma, D. P. and Welling, M. An introduction to variational autoencoders. *Found. Trends Mach. Learn.*, 12 (4):307–392, November 2019. ISSN 1935-8237. doi: 10.1561/22000000056. URL <https://doi.org/10.1561/22000000056>.
- Kingma, D. P. and Welling, M. Auto-encoding variational bayes, 2022. URL <https://arxiv.org/abs/1312.6114>.
- Knight, P. A. The Sinkhorn-Knopp Algorithm: Convergence and Applications. In Frommer, A., Mahoney, M. W., and Szyld, D. B. (eds.), *Web Information Retrieval and Linear Algebra Algorithms*, volume 7071 of *Dagstuhl Seminar Proceedings (DagSemProc)*, pp. 1–18, Dagstuhl, Germany, 2007. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. doi: 10.4230/DagSemProc.07071.16. URL <https://drops.dagstuhl.de/entities/document/10.4230/DagSemProc.07071.16>.
- Lavenant, H., Zhang, S., Kim, Y.-H., Schiebinger, G., et al. Toward a mathematical theory of trajectory inference. *The Annals of Applied Probability*, 34(1A):428–500, 2024.
- Léonard, C. A survey of the schrödinger problem and some of its connections with optimal transport. *Discrete and Continuous Dynamical Systems*, 34(4):1533–1574, 2014. ISSN 1078-0947. doi: 10.3934/dcds.2014.34.1533. URL <https://www.aims sciences.org/article/id/d5bcf817-901d-4104-b7da-eade7847c53e>.
- Mallasto, A., Heinonen, M., and Kaski, S. Bayesian inference for optimal transport with stochastic cost. In Balasubramanian, V. N. and Tsang, I. (eds.), *Proceedings of The 13th Asian Conference on Machine Learning*, volume 157 of *Proceedings of Machine Learning Research*, pp. 1601–1616. PMLR, 17–19 Nov

2021. URL <https://proceedings.mlr.press/v157/mallasto21a.html>.
- Mena, G., Belanger, D., Linderman, S., and Snoek, J. Learning latent permutations with gumbel-sinkhorn networks. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=Byt3oJ-0W>.
- Mohajerin Esfahani, P. and Kuhn, D. Data-driven distributionally robust optimization using the Wasserstein metric: performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1):115–166, sep 2018. doi: 10.1007/s10107-017-1172-1. URL <https://doi.org/10.1007/s10107-017-1172-1>.
- Mukherjee, D., Guha, A., Solomon, J. M., Sun, Y., and Yurochkin, M. Outlier-robust optimal transport. In Meila, M. and Zhang, T. (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 7850–7860. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/mukherjee21a.html>.
- Negri, M. M., Torres, F. A., and Roth, V. Conditional matrix flows for gaussian graphical models. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=GYnbubCXhE>.
- Oquab, M., Darcet, T., Moutakanni, T., Vo, H. V., Szafraniec, M., Khalidov, V., Fernandez, P., HAZIZA, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jegou, H., Mairal, J., Labatut, P., Joulin, A., and Bojanowski, P. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=a68Sut6zFt>. Featured Certification.
- Peyré, G. and Cuturi, M. *Computational Optimal Transport*. 01 2019. ISBN 9781680835519. doi: 10.1561/9781680835519.
- Ranganath, R., Gerrish, S., and Blei, D. Black Box Variational Inference. In Kaski, S. and Corander, J. (eds.), *Proceedings of the Seventeenth International Conference on Artificial Intelligence and Statistics*, volume 33 of *Proceedings of Machine Learning Research*, pp. 814–822, Reykjavik, Iceland, 22–25 Apr 2014. PMLR. URL <https://proceedings.mlr.press/v33/ranganath14.html>.
- Redko, I., Courty, N., Flamary, R., and Tuia, D. Optimal transport for multi-source domain adaptation under target shift. In Chaudhuri, K. and Sugiyama, M. (eds.), *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pp. 849–858. PMLR, 16–18 Apr 2019. URL <https://proceedings.mlr.press/v89/redko19a.html>.
- Rezende, D. and Mohamed, S. Variational inference with normalizing flows. In Bach, F. and Blei, D. (eds.), *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 1530–1538, Lille, France, 07–09 Jul 2015. PMLR. URL <https://proceedings.mlr.press/v37/rezende15.html>.
- Rezende, D. J., Mohamed, S., and Wierstra, D. Stochastic backpropagation and approximate inference in deep generative models. In Xing, E. P. and Jebara, T. (eds.), *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pp. 1278–1286, Beijing, China, 22–24 Jun 2014. PMLR. URL <https://proceedings.mlr.press/v32/rezende14.html>.
- Rosen, O., Jiang, W., King, G., and Tanner, M. Bayesian and frequentist inference for ecological inference: The rxc case. *Statistica Neerlandica*, 55:134 – 156, 12 2001. doi: 10.1111/1467-9574.00162.
- Saenko, K., Kulis, B., Fritz, M., and Darrell, T. Adapting visual category models to new domains. In *Proceedings of the 11th European Conference on Computer Vision: Part IV, ECCV’10*, pp. 213–226, Berlin, Heidelberg, 2010. Springer-Verlag. ISBN 364215560X.
- Schiebinger, G., Shu, J., Tabaka, M., Cleary, B., Subramanian, V., Solomon, A., Gould, J., Liu, S., Lin, S., Berube, P., Lee, L., Chen, J., Brumbaugh, J., Rigollet, P., Hochedlinger, K., Jaenisch, R., Regev, A., and Lander, E. S. Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming. *Cell*, 176(4):928–943.e22, 2019. ISSN 0092-8674. doi: <https://doi.org/10.1016/j.cell.2019.01.006>. URL <https://www.sciencedirect.com/science/article/pii/S009286741930039X>.
- Sinha, A., Namkoong, H., and Duchi, J. Certifiable distributional robustness with principled adversarial training. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=Hk6kPgZA->.
- Sinkhorn, R. A relationship between arbitrary positive matrices and doubly stochastic matrices. *The An-*

- nals of Mathematical Statistics*, 35(2):876–879, 1964. ISSN 00034851. URL <http://www.jstor.org/stable/2238545>.
- Stromme, A. Sampling from a schrödinger bridge. In Ruiz, F., Dy, J., and van de Meent, J.-W. (eds.), *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 4058–4067. PMLR, 25–27 Apr 2023. URL <https://proceedings.mlr.press/v206/stromme23a.html>.
- Tay, Y., Bahri, D., Yang, L., Metzler, D., and Juan, D.-C. Sparse Sinkhorn attention. In III, H. D. and Singh, A. (eds.), *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 9438–9447. PMLR, 13–18 Jul 2020. URL <https://proceedings.mlr.press/v119/tay20a.html>.
- Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., and Jegou, H. Training data-efficient image transformers & distillation through attention. In Meila, M. and Zhang, T. (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 10347–10357. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/touvron21a.html>.
- Ubaru, S., Chen, J., and Saad, Y. Fast estimation of $\text{tr}(f(a))$ via stochastic lanczos quadrature. *SIAM Journal on Matrix Analysis and Applications*, 38(4):1075–1099, 2017.
- Uppal, A., Stensbo-Smidt, K., Boomsma, W., and Frellsen, J. Implicit variational inference for high-dimensional posteriors. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=Sxu7xlUJGx>.
- van der Linden, P. A., García-Castellanos, A., Vadgama, S., Kuipers, T. P., and Bekkers, E. J. Learning symmetries via weight-sharing with doubly stochastic tensors. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C. (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 42738–42764. Curran Associates, Inc., 2024. doi: 10.52202/079017-1354. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/4b34143d8a1e651f00002e31b3b72e95-Paper-Conference.pdf.
- Woodard, D., Schmidler, S., and Huber, M. Conditions for rapid mixing of parallel and simulated tempering on multimodal distributions. *The Annals of Applied Probability*, 19, 06 2009. doi: 10.1214/08-AAP555.
- Yang, Y. and Gao, J. mHC-lite: You don’t need 20 sinkhorn-knopp iterations. In *ICLR 2026 Workshop on Geometry-grounded Representation Learning and Generative Modeling*, 2026. URL <https://openreview.net/forum?id=5IJX6kvOif>.
- Zhang, K., Zheng, V., Wang, Q., Kwok, J., Yang, Q., and Marsic, I. Covariate shift in hilbert space: A solution via surrogate kernels. In Dasgupta, S. and McAllester, D. (eds.), *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pp. 388–395, Atlanta, Georgia, USA, 17–19 Jun 2013. PMLR. URL <https://proceedings.mlr.press/v28/zhang13b.html>.
- Zhang, Z., Li, T., and Zhou, P. Learning stochastic dynamics from snapshots through regularized unbalanced optimal transport. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=gQlxd3Mtru>.

A. Truncated Sinkhorn Scaling

This appendix formalizes the use of a finite Sinkhorn iteration budget T in our variational family and establishes uniform geometric convergence of the truncated coupling map.

Setup. Fix the compact latent set $Z := [-c, c]^r$ and define the gauge-fixed logit map $B(z) := \text{mat}(Qz) \in \mathcal{G}^\perp$, where $Q \in \mathbb{R}^{nm \times r}$ has orthonormal columns spanning $\text{vec}(\mathcal{G}^\perp)$. Let

$$K(z) := \exp(B(z)) \in \mathbb{R}_{++}^{n \times m}, \quad \Psi(z) := \mathcal{S}(K(z); \mu, \nu) \in \mathcal{M},$$

denote the exact Sinkhorn coupling. Starting from $u^0 = \mathbf{1}_n$, define the truncated iterates

$$v^t(z) := \nu \oslash (K(z)^\top u^{t-1}(z)), \quad u^t(z) := \mu \oslash (K(z)v^t(z)), \quad \Psi_T(z) := \text{diag}(u^T(z)) K(z) \text{diag}(v^T(z)). \quad (14)$$

We measure marginal feasibility via the residual vector

$$\text{res}_T(z) := \begin{bmatrix} \Psi_T(z) \mathbf{1}_m - \mu \\ \Psi_T(z)^\top \mathbf{1}_n - \nu \end{bmatrix} \in \mathbb{R}^{n+m}. \quad (15)$$

We first establish the following structural assumptions:

- A1.** The latent set Z is compact and Q is a fixed linear operator, ensuring a finite bound $b := \sup_{z \in Z} \|B(z)\|_\infty$.
- A2.** The energy E extends to a Lipschitz function on a compact set $\mathcal{U} \subset \mathbb{R}_{++}^{n \times m}$ that contains $\{\Psi(z) : z \in Z\}$ and $\{\Psi_T(z) : z \in Z\}$ (for the fixed truncation T used in the objective).

Under **A1**, $K(z) = \exp(B(z))$ is C^2 in z , with uniformly bounded first derivatives on Z , and all kernels satisfy $e^{-b} \leq K_{ij}(z) \leq e^b$ for all (i, j) and $z \in Z$. We define the Hilbert projective metric for $x, y \in \mathbb{R}_{++}^d$ as $d_H(x, y) := \log(\max_i x_i/y_i) - \log(\min_i x_i/y_i)$. For $K \in \mathbb{R}_{++}^{n \times m}$ define the projective diameter $\Delta(K)$ and contraction coefficient $\kappa(K)$ as

$$\Delta(K) := \log \sup_{i, i', j, j'} \frac{K_{ij} K_{i'j'}}{K_{ij'} K_{i'j}}, \quad \kappa(K) := \tanh(\Delta(K)/4).$$

Hence, **A1** implies the projective diameter satisfies $\Delta(K(z)) \leq 4b$ because:

$$\frac{K_{ij} K_{i'j'}}{K_{ij'} K_{i'j}} \leq \frac{e^b e^b}{e^{-b} e^{-b}} = e^{4b}.$$

Theorem A.1. Assume **A1**. Then there exists a rate

$$\rho := \sup_{z \in Z} \kappa(K(z))^2 \leq \tanh(b)^2 < 1$$

and a constant $C_\Gamma < \infty$ (depending only on b, μ, ν) such that for all $z \in Z$ and $T \geq 1$,

$$\|\Psi_T(z) - \Psi(z)\|_1 \leq C_\Gamma \rho^{T-1}, \quad \|\text{res}_T(z)\|_1 \leq C_\Gamma \rho^{T-1}. \quad (16)$$

Proof. The Birkhoff–Hopf theorem implies that the projective iterates contract: the full Sinkhorn step (one v update followed by one u update) is a contraction in d_H with coefficient at most $\kappa(K(z))^2$ (Franklin & Lorenz, 1989; Idel, 2016). Under **A1**, this yields a uniform rate $\rho < 1$ on Z . To translate contraction in d_H into an ℓ^1 bound for Ψ_T , we use that the iterates (u^t, v^t) remain in a forward-invariant, projectively bounded set determined by (b, μ, ν) (e.g., $d_H(u^t, \mathbf{1}_n)$ is uniformly bounded). On such a compact log-slice, d_H controls $\|\log u - \log(\alpha u)\|_\infty$ for an appropriate scalar α . Since $(u, v) \mapsto \text{diag}(u) K \text{diag}(v)$ is smooth and hence Lipschitz on this compact set, the contraction transfers to $\|\Psi_T - \Psi\|_1$ with a uniform constant C_Γ . Note that by construction $\Psi_T(z) \mathbf{1}_m = \mu$, hence the residual reduces to the column error $\|\Psi_T(z)^\top \mathbf{1}_n - \nu\|_1$, which decays geometrically under the same projective contraction. $\square$

Now according to **A2**, let $E : \mathcal{M} \rightarrow \mathbb{R}$ be Lipschitz on the compact subset $\{\Psi(z) : z \in Z\} \subset \mathcal{M}$. Then Theorem A.1 implies

$$\sup_{z \in Z} |E(\Psi_T(z)) - E(\Psi(z))| \leq \text{Lip}(E) \sup_{z \in Z} \|\Psi_T(z) - \Psi(z)\|_1 = \mathcal{O}(\rho^T).$$

Lemma A.2. Suppose $\Phi : Z \times \mathcal{X} \rightarrow \mathcal{X}$ is C^1 and for all $z \in Z$, $x \in \mathcal{X}$, $\|D_x \Phi_z(x)\| \leq \eta < 1$ and $\sup_{z,x} \|D_z \Phi_z(x)\| < \infty$. Let $x^t(z)$ be defined by $x^t(z) = \Phi_z(x^{t-1}(z))$ with x^0 independent of z . Then the fixed point $x^*(z)$ is C^1 , and the derivative iterates $Dx^t(z)$ converge to $Dx^*(z)$ geometrically and uniformly:

$$\sup_{z \in Z} \|Dx^t(z) - Dx^*(z)\| \leq C \eta^t,$$

where C depends on $\sup \|D_z \Phi_z\|$ and $(1 - \eta)^{-1}$.

We now address convergence of the Jacobian

$$J_T(z) := \nabla_z \text{vec}(\Psi_T(z)) \in \mathbb{R}^{nm \times r}, \quad J_\infty(z) := \nabla_z \text{vec}(\Psi(z)).$$

Uniform geometric convergence of J_T is not implied by Theorem A.1 alone; it additionally requires stability of the *linearization* of Sinkhorn under truncation. We state a sufficient condition that can be verified under bounded logits and strictly positive marginals.

Let $\alpha^t(z) := \log u^t(z) \in \mathbb{R}^n$ and $\beta^t(z) := \log v^t(z) \in \mathbb{R}^m$. Fix the 1-dimensional scaling ambiguity by imposing a gauge, e.g. $\langle \mu, \alpha \rangle = 0$ at every step. Let $x^t(z)$ denote the resulting gauge-fixed log-scaling state (e.g. $x^t = \alpha^t$). Write the full Sinkhorn step as a map $x^t = \Phi_z(x^{t-1})$.

A3. There exists a norm $\|\cdot\|$ on the gauge-fixed state space, a compact forward-invariant set $\mathcal{X}$, and constants $\eta \in (0, 1)$ and $L_z < \infty$ such that for all $z \in Z$ and all $x \in \mathcal{X}$:

$$\|D_x \Phi_z(x)\| \leq \eta, \quad \|D_z \Phi_z(x)\| \leq L_z.$$

Moreover, Φ_z is C^1 jointly in (x, z) on $\mathcal{X} \times Z$.

Theorem A.3. Assume **A1** and **A3**. Then: (i) the exact map $\Psi : Z \rightarrow \mathcal{M}$ is C^1 , (ii) the truncated Jacobians converge uniformly at a geometric rate: there exist constants $C_J < \infty$ and $\tilde{\rho} \in (0, 1)$ such that for all $T \geq 1$,

$$\sup_{z \in Z} \|J_T(z) - J_\infty(z)\| \leq C_J \tilde{\rho}^{T-1}, \quad (17)$$

where $\tilde{\rho}$ can be taken as η (up to norm equivalence constants induced by the gauge).

Proof. Under **A3**, the fixed point $x^*(z)$ of $x = \Phi_z(x)$ exists uniquely on $\mathcal{X}$ and depends smoothly on z by the implicit function theorem / contraction mapping theorem for C^1 contractions. Differentiating the recursion yields

$$D_z x^t(z) = D_x \Phi_z(x^{t-1}(z)) D_z x^{t-1}(z) + D_z \Phi_z(x^{t-1}(z)).$$

Subtracting the fixed-point derivative equation and using $\|D_x \Phi_z\| \leq \eta < 1$ gives a linear contraction in the derivative error, hence $\sup_z \|D_z x^t(z) - D_z x^*(z)\| = \mathcal{O}(\eta^t)$ uniformly. Finally, $\Psi_T(z)$ is a smooth composition of $(u^T(z), v^T(z), K(z))$ with $(u, v, K) \mapsto \text{diag}(u)K\text{diag}(v)$, so convergence of derivatives of the scaling state implies (17). $\square$

B. Implementation details

The following implementation details are the same across all experiments unless stated otherwise.

Construction of Q . We do not form $Q \in \mathbb{R}^{nm \times r}$ but implement $B = \text{mat}(Qz) \in \mathcal{G}^\perp$ via an explicit factorization. We precompute the QR bases

$$\begin{aligned} A_n &= [e_1 - e_n, \dots, e_{n-1} - e_n] = UR_n, \\ A_m &= [e_1 - e_m, \dots, e_{m-1} - e_m] = VR_m, \end{aligned}$$

so $U \in \mathbb{R}^{n \times (n-1)}$, $V \in \mathbb{R}^{m \times (m-1)}$ have orthonormal columns and satisfy $U^\top \mathbf{1}_n = 0$, $V^\top \mathbf{1}_m = 0$. We can now choose any $z \in \mathbb{R}^r$, reshaped to form $S = \text{reshape}(z, (n-1, m-1))$ and set $B := USV^\top$, which enforces $B\mathbf{1}_m = 0$ and $B^\top \mathbf{1}_n = 0$ by construction. This is equivalent to using $Q = V \otimes U$ but avoids storing Q .

Hyperparameters. We fix the number of Sinkhorn iterations T such that the column marginal residual is below 10^{-10} for a reasonable n . Typically, this happens within $T \leq 20$ steps for $n \leq 300$. We use $S = 4$ Monte Carlo samples for the expectation over ξ . For the trace estimator of the Gram logarithm, we use $M = 8$ Rademacher probe vectors and $L = 10$ Lanczos steps. To choose c , we use the robust lower-bound heuristic $c \geq \frac{\max(C) - \min(C)}{2\varepsilon}$ and set $c = 1.2 \times \frac{\max(C) - \min(C)}{2\varepsilon}$. To ensure numerical stability at larger scales, all Sinkhorn operations and variational density evaluations were performed in the log-domain using the log-sum-exp trick.

Optimization details. We use Adam with learning rate $\eta = 10^{-3}$, $(\beta_1, \beta_2) = (0.9, 0.999)$, and $\epsilon_{\text{ADAM}} = 10^{-8}$. We apply gradient clipping with maximum global norm 1.0 to stabilize training. We use a linear warmup for the first T_{warm} iterations up to the base learning rate, followed by cosine decay down to a minimum learning rate of 10^{-5} :

$$\eta_t = \begin{cases} \eta \cdot \frac{t}{T_{\text{warm}}}, & t \leq T_{\text{warm}}, \\ \eta_{\min} + \frac{1}{2}(\eta - \eta_{\min}) \left(1 + \cos \left(\pi \frac{t - T_{\text{warm}}}{T_{\text{max}} - T_{\text{warm}}} \right) \right), & T_{\text{warm}} < t \leq T_{\text{max}}, \end{cases}$$

with $\eta_{\min} = 10^{-5}$, $T_{\text{max}} = 200$ the total number of training iterations, and $T_{\text{warm}} = 20$. We use early stopping and terminate training if the ELBO does not improve for 50 consecutive steps.

B.1. Domain Adaptation on Toy Datasets

We utilize the MNIST and USPS handwritten digit datasets. All images are resized to 16×16 pixels and pixel values are normalized to $[0, 1]$. For both experiments, we randomly sample $n = 25$ source images and $m = 25$ target images. The variational distribution is parameterized by a RealNVP normalizing flow consisting of 2 coupling layers. Within each coupling layer, the translation and scale networks are implemented as 2-layer MLPs with 32 hidden units and ReLU activations.

B.2. High-Dimensional Scalability

We generate synthetic 2D point clouds drawn from Gaussian mixtures. The base cost matrix C is the squared Euclidean distance. We simulate cost uncertainty by generating $N = 20$ positive noisy realizations

$$C_{ij}^{(k)} = C_{ij} \exp \left(\sigma Z_{ij}^{(k)} - \frac{1}{2} \sigma^2 \right), \quad Z_{ij}^{(k)} \sim \mathcal{N}(0, 1),$$

with noise level $\sigma = 0.1$. The centering term keeps $\mathbb{E}[C_{ij}^{(k)} \mid C_{ij}] = C_{ij}$ while preserving nonnegativity. We compare SPVI against the BayesOT implementation (Mallasto et al., 2021) using the No-U-Turn Sampler (NUTS) (Hoffman & Gelman, 2014). Chains are warm-started using the deterministic Sinkhorn solution on the mean cost matrix. For the HMC runtime benchmark ($n \leq 256$), we use scale-dependent NUTS budgets: 50 burn-in/100 samples at $n = 32$, 100/200 at $n = 64$, 200/500 at $n = 128$, and 500/1000 at $n = 256$. We average over 10 runs with a target acceptance rate of 0.8. Chain convergence was assessed using the $\hat{R}$ diagnostic, with a threshold of $\hat{R} < 1.1$ for all parameters and a 10-hour wall-clock limit. As the dimensionality of the transport polytope grows quadratically with the support size, we scale the capacity of the architecture and the HMC sampling budget accordingly, as shown in Table 4. The conditioner for SPVI uses 2 residual conv blocks and 3×3 kernels for all cases.

Posterior sample distance. Let $\{\Gamma_i^{\text{VI}}\}_{i=1}^S$ and $\{\Gamma_j^{\text{HMC}}\}_{j=1}^S$ denote equal-size posterior sample sets, and let $g_i^{\text{VI}} = \text{vec}(\Gamma_i^{\text{VI}})$ and $g_j^{\text{HMC}} = \text{vec}(\Gamma_j^{\text{HMC}})$. For $n \leq 128$, we report the empirical 2-Wasserstein distance

$$\widehat{\mathcal{W}}_2 = \left[\min_{\pi \in \Pi_S} \frac{1}{S} \sum_{i=1}^S \|g_i^{\text{VI}} - g_{\pi(i)}^{\text{HMC}}\|_2^2 \right]^{1/2},$$

where Π_S is the set of permutations of $\{1, \dots, S\}$ with $S = 200$. For $n > 128$, exact empirical matching is expensive because $g_i \in \mathbb{R}^{n^2}$, so we report the sliced 2-Wasserstein distance

$$\widehat{\mathcal{S}\mathcal{W}}_2 = \left[\frac{1}{K} \sum_{k=1}^K \mathcal{W}_2^2 \left(\frac{1}{S} \sum_{i=1}^S \delta_{\theta_k^\top g_i^{\text{VI}}}, \frac{1}{S} \sum_{j=1}^S \delta_{\theta_k^\top g_j^{\text{HMC}}} \right) \right]^{1/2},$$

with random directions $\theta_k \sim \text{Unif}(\mathbb{S}^{n^2-1})$. In one dimension, each inner $\mathcal{W}_2$ is computed by sorting the projected samples.

Table 3. Posterior quality diagnostics for SPVI in the high-dimensional scalability experiment. Metrics are averaged over 5 runs. Variance is computed using 200 samples.

Size (n)	$r = (n - 1)^2$	$\mathcal{W}_2 / S\mathcal{W}_2 \downarrow$	$\text{Var}[\langle C, \Gamma \rangle]$
$n = 32$	961	0.00248 ± 0.00060	0.00169 ± 0.00009
$n = 64$	3,969	0.00261 ± 0.00055	0.00856 ± 0.00033
$n = 128$	16,129	0.00169 ± 0.00041	0.00930 ± 0.00042
$n = 256$	65,025	0.00218 ± 0.00044	0.00820 ± 0.00061
$n = 512$	261,121	—	0.00861 ± 0.00015

Table 4. SPVI architecture and HMC hyperparameters across scales.

n	r	Layers	Hidden channels	Burn-in	NUTS samples
32	961	2	16	50	100
64	3,969	4	32	100	200
128	16,129	4	32	200	500
256	65,025	8	32	500	1,000
512	261,121	8	32	1,000	2,000

B.3. Joint Representation and Transport Learning

We utilize an ensemble of four frozen pre-trained encoders to generate diverse cost views. For each encoder, the input images are resized to 224×224 and normalized using ImageNet statistics. For each view k , we compute pairwise distances between source and target embeddings:

$$D_{ij}^{(k)} = d(r^{(k)}(x_i^s), r^{(k)}(x_j^t)).$$

where

$$d(u, v) = \|u - v\|^2.$$

Distances from different encoders have very different scales, so we normalize each view:

$$\tilde{D}_{ij}^{(k)} = \frac{D_{ij}^{(k)}}{\text{median}(D^{(k)}) + \epsilon}.$$

The resulting $\tilde{D}^{(k)}$ are comparable across k . We solve the block coordinate descent problem using $T_{\text{outer}} = 30$ iterations and $T = 40$ for Sinkhorn. We use the Adam optimizer for both ϕ and ψ with learning rates $\eta_\phi = 10^{-3}$ and $\eta_\psi = 10^{-3}$, respectively. We search for the best $\alpha \in \{10^{-5}, 10^{-4}, 10^{-3}, \dots, 1\}$, $\lambda_s \in \{10^{-3}, \dots, 1, 1.5\}$, and weight decay $\lambda_\Omega \in \{10^{-4}, \dots, 1\}$. The label consistency weight $\eta(t)$ ramps linearly from 0 to 1 over $T_{\text{ramp}} = 0.5 \times T_{\text{outer}}$. We pick a single classifier feature map $\varphi(x)$, then

$$g_\psi(x) = W\varphi(x) + b, \quad \psi = (W, b), \quad W \in \mathbb{R}^{K \times d}, \quad b \in \mathbb{R}^K,$$

and $g_{\psi,c}(x) = w_c^\top \varphi(x) + b_c$. The squared hinge then uses these margins directly:

$$\ell_{\text{sh}}(Y_{ic}^s, g_{\psi,c}(x)) = \max(0, 1 - Y_{ic}^s g_{\psi,c}(x))^2.$$

Because the label term $L_{ij}(\psi)$ is **shared across views** k , g_ψ should *not* depend on k . To define one shared $\varphi(x)$ from the same frozen encoders, we concatenate their normalized embeddings:

Let $r^{(k)}(x)$ be the frozen embedding from encoder k , and $\bar{r}^{(k)}(x) = r^{(k)}(x) / |r^{(k)}(x)|_2$. Define

$$\varphi(x) = [\bar{r}^{(1)}(x) \parallel \bar{r}^{(2)}(x) \parallel \bar{r}^{(3)}(x) \parallel \bar{r}^{(4)}(x)] \in \mathbb{R}^{d_1+d_2+d_3+d_4}.$$

We warm start the classifier for 5 epochs. For SurK, we search $d \in \{1, 4, 7, \dots, 31\}$. For OT-MM, we search the regularization parameter $\eta \in \{1, \dots, 10^5\}$. We use a linear-kernel SVM for classification for each method.

B.4. Single-Cell Fate Dynamics

We use the Dentate Gyrus scRNA-seq dataset from [Hochgerner et al. \(2018\)](#). Cells are represented by the top 30 principal components ($d = 30$) of the top 2,000 highly variable genes. We sample $n = 300$ cells from the earliest latent-time quantile ($\leq 15\%$ of the latent time) as source and $m = 300$ cells from the latest latent-time quantile ($\geq 85\%$ of the latent time) as target. To construct the cost ensemble $\mathcal{C}$, we generate $N = 50$ subsampling realizations of the inferred developmental dynamics. For each realization, we subsample 80% of the highly variable genes without replacement, recompute the 30-dimensional PCA embedding, and construct a 30-nearest-neighbor graph in the resulting cell-state space. We orient the local transition structure forward along latent time and compute source-to-target transition costs from the resulting forward developmental graph. We compute L -step forward diffusion probabilities $P_{ij}^{(k)}$ from source cells to target cells and define $C_{ij}^{(k)} = -\log(P_{ij}^{(k)} + \delta)$, with $\delta = 10^{-8}$. Edges are reweighted by a forward-time kernel and row-normalized to obtain a Markov transition matrix. This gives an ensemble of plausible transition cost matrices $\mathcal{C} = \{C^{(1)}, \dots, C^{(50)}\}$, reflecting uncertainty in the inferred developmental dynamics rather than perturbations of the transport plans themselves.

The variational posterior $q_\phi(\Gamma|\mathcal{C})$ is parameterized by an 8-layer RealNVP flow. The plasticity score V_i is computed using $S = 200$ posterior samples from the trained flow. For each posterior sample $\Gamma^{(s)}$, we compute the barycentric fate prediction

$$\mathcal{T}_i(\Gamma^{(s)}) = \sum_{j=1}^m \frac{\Gamma_{ij}^{(s)}}{\mu_i} y_j \in \mathbb{R}^{30}.$$

The plasticity score is then estimated as

$$V_i = \frac{1}{S-1} \sum_{s=1}^S \sum_{\ell=1}^{30} \left([\mathcal{T}_i(\Gamma^{(s)})]_\ell - \overline{[\mathcal{T}_i(\Gamma)]}_\ell \right)^2,$$

where

$$\overline{[\mathcal{T}_i(\Gamma)]}_\ell = \frac{1}{S} \sum_{s=1}^S [\mathcal{T}_i(\Gamma^{(s)})]_\ell.$$

C. Impact of Sinkhorn Iterations

For most experiments, we find that $T \leq 20$ results in a marginal violation of 10^{-10} . To isolate the contributions of our architectural choices, we conduct controlled experiments focusing on the Sinkhorn depth, approximation fidelity, and the expressivity of the variational family. Throughout these trials, we fix the entropic regularization $\epsilon = 0.05$, the problem size $n = 256$, and the noise level $\sigma = 0.1$ to ensure consistency across comparisons. We vary $T \in \{5, 10, 20, 50\}$ while maintaining a fixed budget of $S = 4$ Monte Carlo samples, $L = 10$ Lanczos steps, and $M = 8$ probes over 200 iterations. Results are averaged over 5 runs.

D. Proof of Proposition 4.2

Proposition D.1 (Global Diffeomorphism of the Gauge-Fixed Map). *Let $\mathcal{G} = \{a\mathbf{1}_m^\top + \mathbf{1}_n b^\top : a \in \mathbb{R}^n, b \in \mathbb{R}^m\}$ be the subspace of additive gauge freedoms, and let $\mathcal{G}^\perp$ be its orthogonal complement in $\mathbb{R}^{n \times m}$. Let $Q \in \mathbb{R}^{nm \times r}$ be a matrix with orthonormal columns spanning $\text{vec}(\mathcal{G}^\perp)$. Define the map $\Psi : \mathbb{R}^r \rightarrow \mathcal{M}$ by*

$$\Psi(z) := \mathcal{S}(\exp(\text{mat}(Qz)); \mu, \nu). \quad (18)$$

Then Ψ is a smooth global diffeomorphism from $\mathbb{R}^r$ onto $\mathcal{M}$. In particular, it is globally injective and its Jacobian $J_\Psi(z)$ has full column rank r everywhere.

Proof. Let $\mathcal{S}_{\log} : \mathbb{R}^{n \times m} \rightarrow \mathcal{M}$ denote the log-domain Sinkhorn map $\mathcal{S}_{\log}(B) = \mathcal{S}(\exp(B); \mu, \nu)$. By the uniqueness of Sinkhorn scaling factors ([Peyré & Cuturi, 2019](#)), for any $\Gamma \in \mathcal{M}$, the kernel $K = \exp(B)$ such that $\mathcal{S}(K) = \Gamma$ is unique up to left and right diagonal scaling. In the log-domain, this implies $\mathcal{S}_{\log}(B) = \mathcal{S}_{\log}(B')$ if and only if $B - B' \in \mathcal{G}$, where $\mathcal{G}$ is the subspace of gauge directions. Thus, the fibers are affine subspaces:

$$\mathcal{S}_{\log}^{-1}(\Gamma) = \log \Gamma + \mathcal{G}. \quad (19)$$

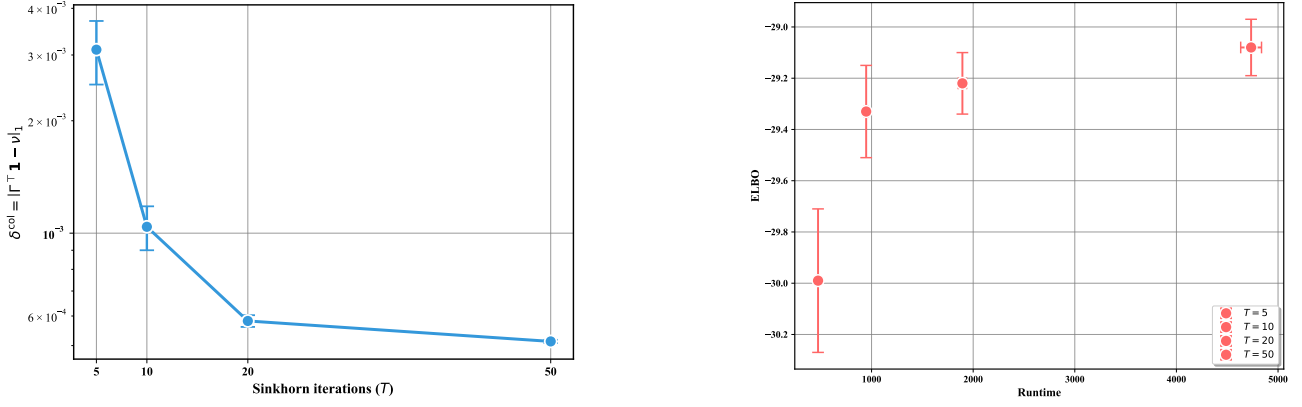


Figure 7. (a) Column error vs. Sinkhorn iterations T shows approximately geometric decay until hitting a precision floor near $T = 50$. (b) ELBO vs. runtime for different Sinkhorn depths T . Higher T improves performance at increased computational cost.

The domain of Ψ is the subspace $\mathcal{G}^\perp = \text{im}(Q)$. Since $\mathbb{R}^{n \times m} = \mathcal{G} \oplus \mathcal{G}^\perp$ is an orthogonal direct sum, every affine translate of $\mathcal{G}$ intersects $\mathcal{G}^\perp$ at exactly one point. Specifically, for any $\Gamma \in \mathcal{M}$, the unique $z \in \mathbb{R}^r$ satisfying $\Psi(z) = \Gamma$ is given by the orthogonal projection of $\log \Gamma$ onto $\mathcal{G}^\perp$ expressed in the Q -basis:

$$\Psi^{-1}(\Gamma) = Q^\top \text{vec}(P_{\mathcal{G}^\perp}(\log \Gamma)). \quad (20)$$

The existence of this explicit inverse map $\Psi^{-1} : \mathcal{M} \rightarrow \mathbb{R}^r$ establishes global bijectivity. The map Ψ is a composition of smooth maps: the linear embedding Q , the entrywise exponential, and the Sinkhorn operator. The Sinkhorn operator $\mathcal{S}(\cdot; \mu, \nu)$ is smooth on the positive orthant $\mathbb{R}_{++}^{n \times m}$ as a consequence of the Implicit Function Theorem applied to the marginal constraints (Knight, 2007). Similarly, Ψ^{-1} is smooth on $\mathcal{M}$ as it is a composition of the smooth entrywise logarithm and linear projections. Since Ψ and Ψ^{-1} are both smooth, Ψ is a global diffeomorphism. Consequently, the differential $d\Psi_z$ is an isomorphism between the tangent spaces $T_z \mathbb{R}^r$ and $T_{\Psi(z)} \mathcal{M}$, ensuring that $J_\Psi(z)$ has full column rank r for all $z \in \mathbb{R}^r$. $\square$